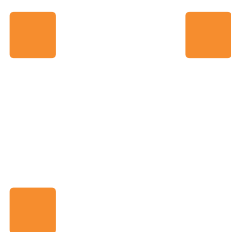


Ensemble of tree-based models

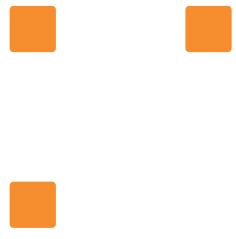


Combine many decision trees into powerful models!

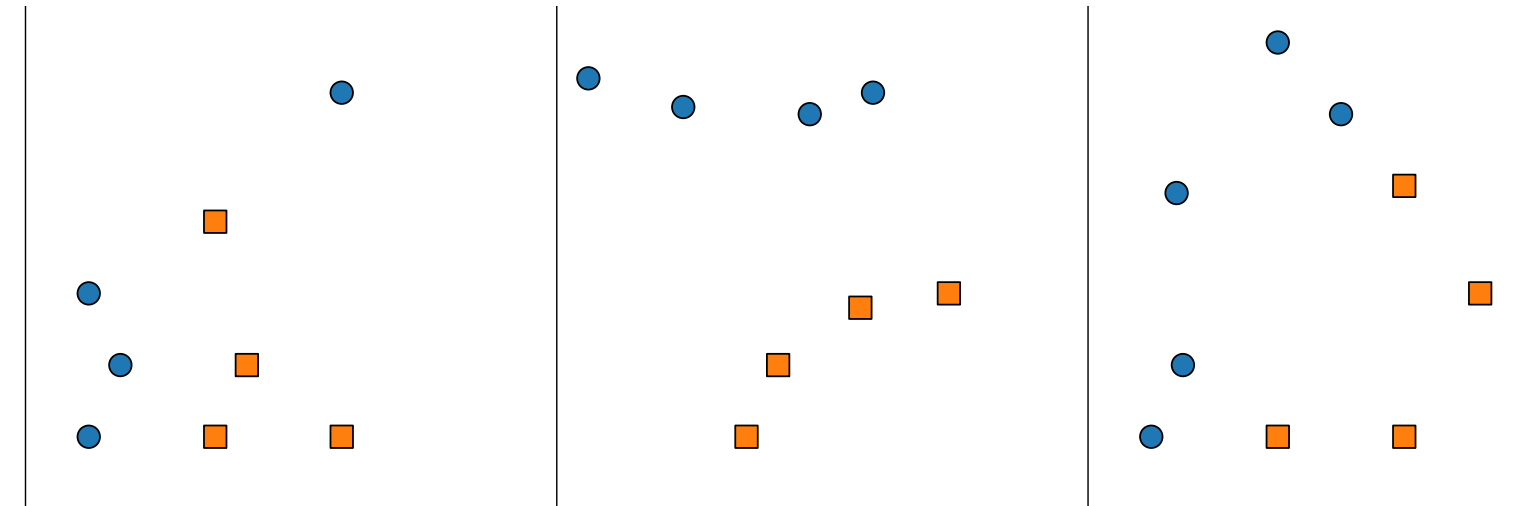
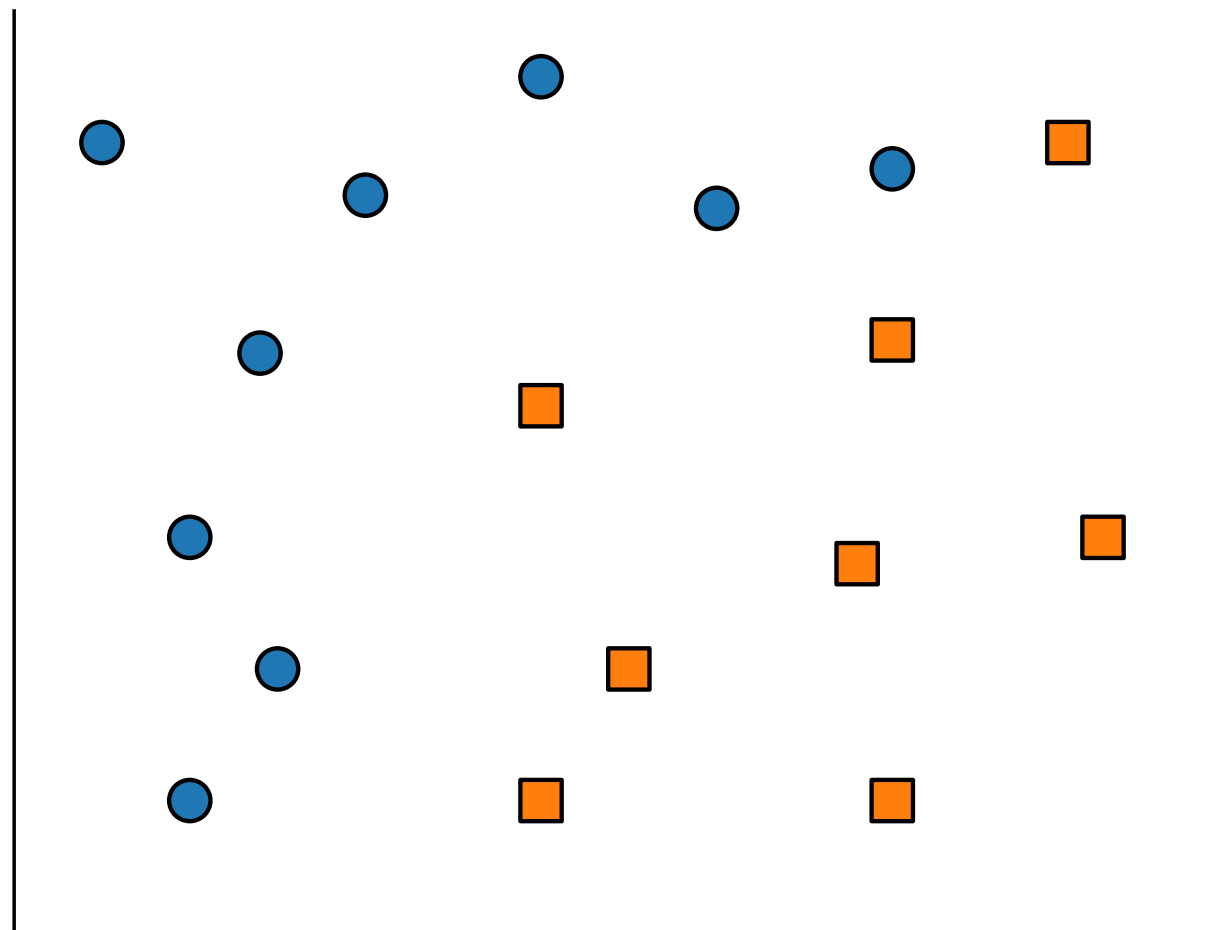


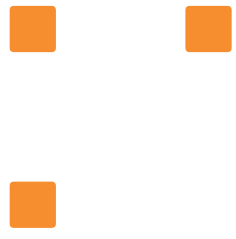
Part 1

Bagging and Random Forests

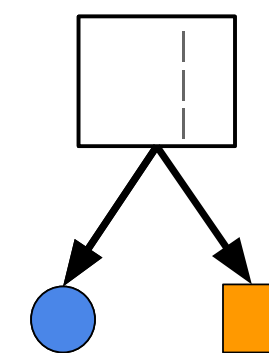
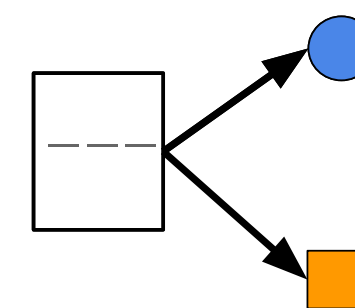
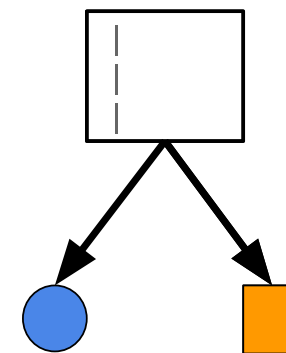
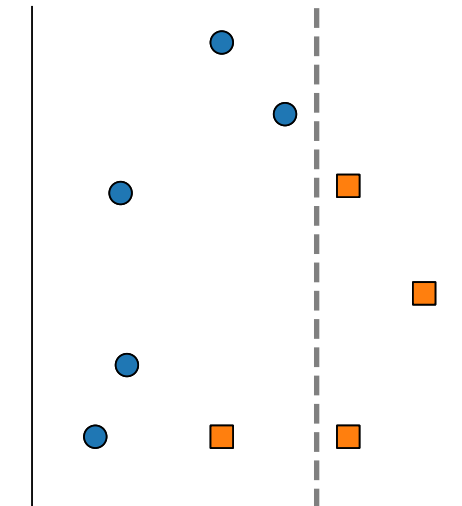
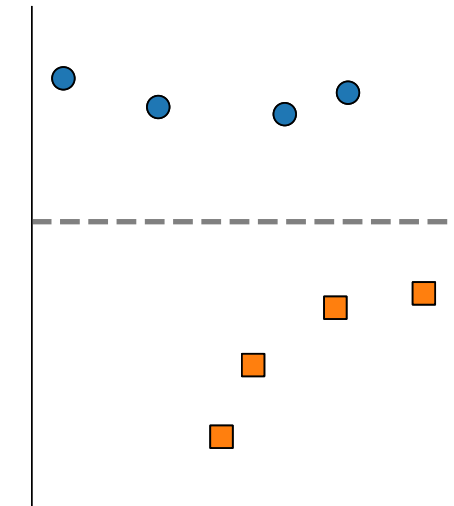
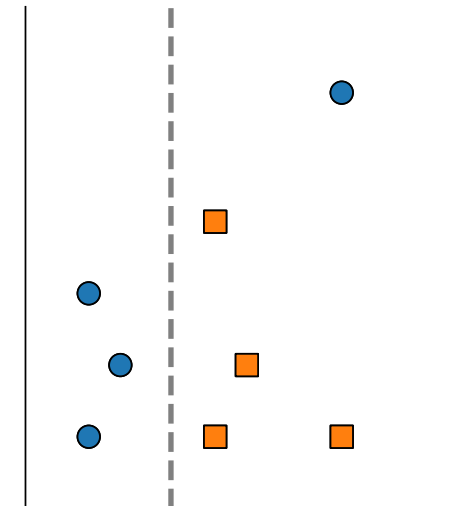
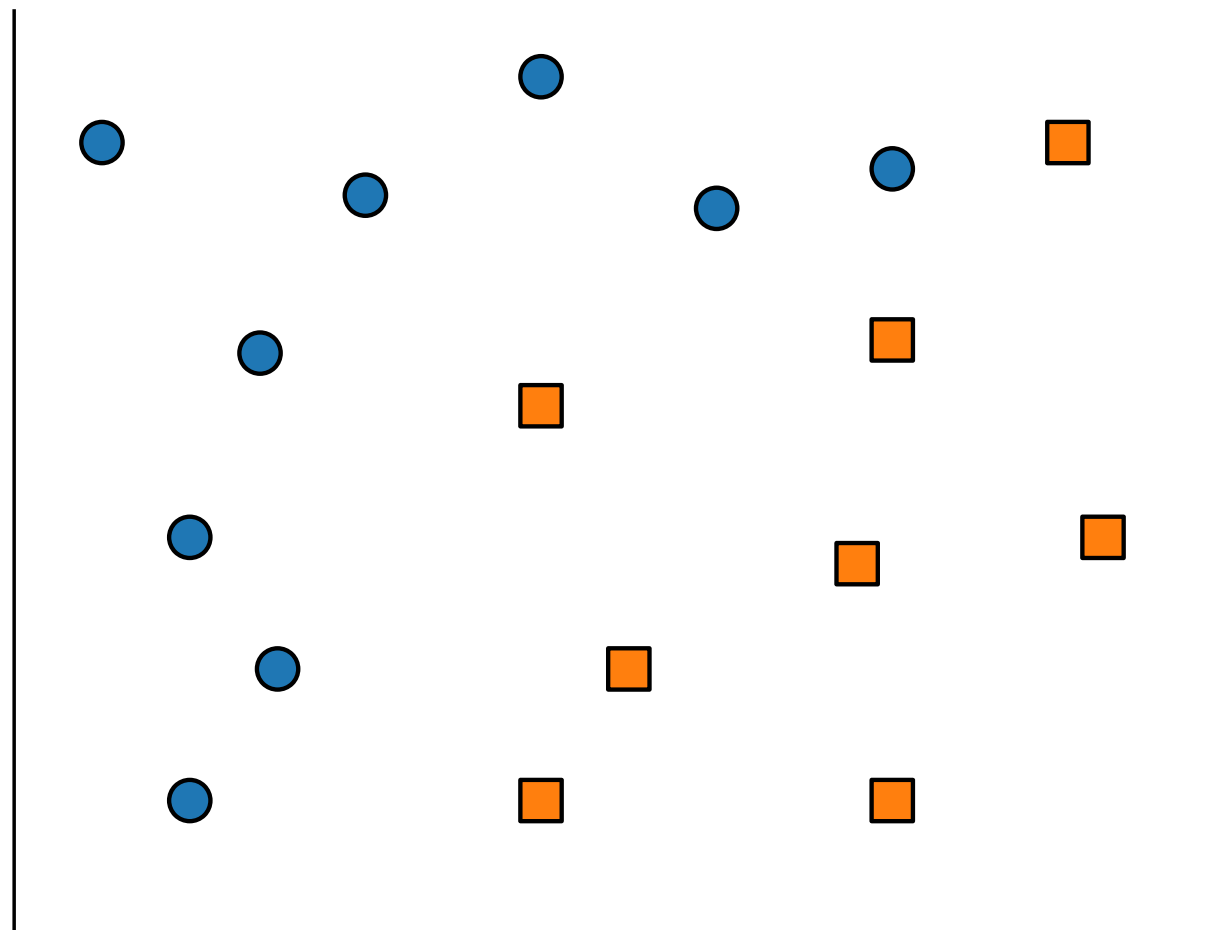


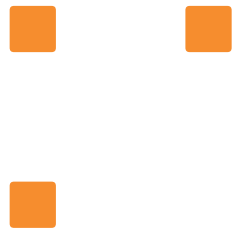
Bagging for classification



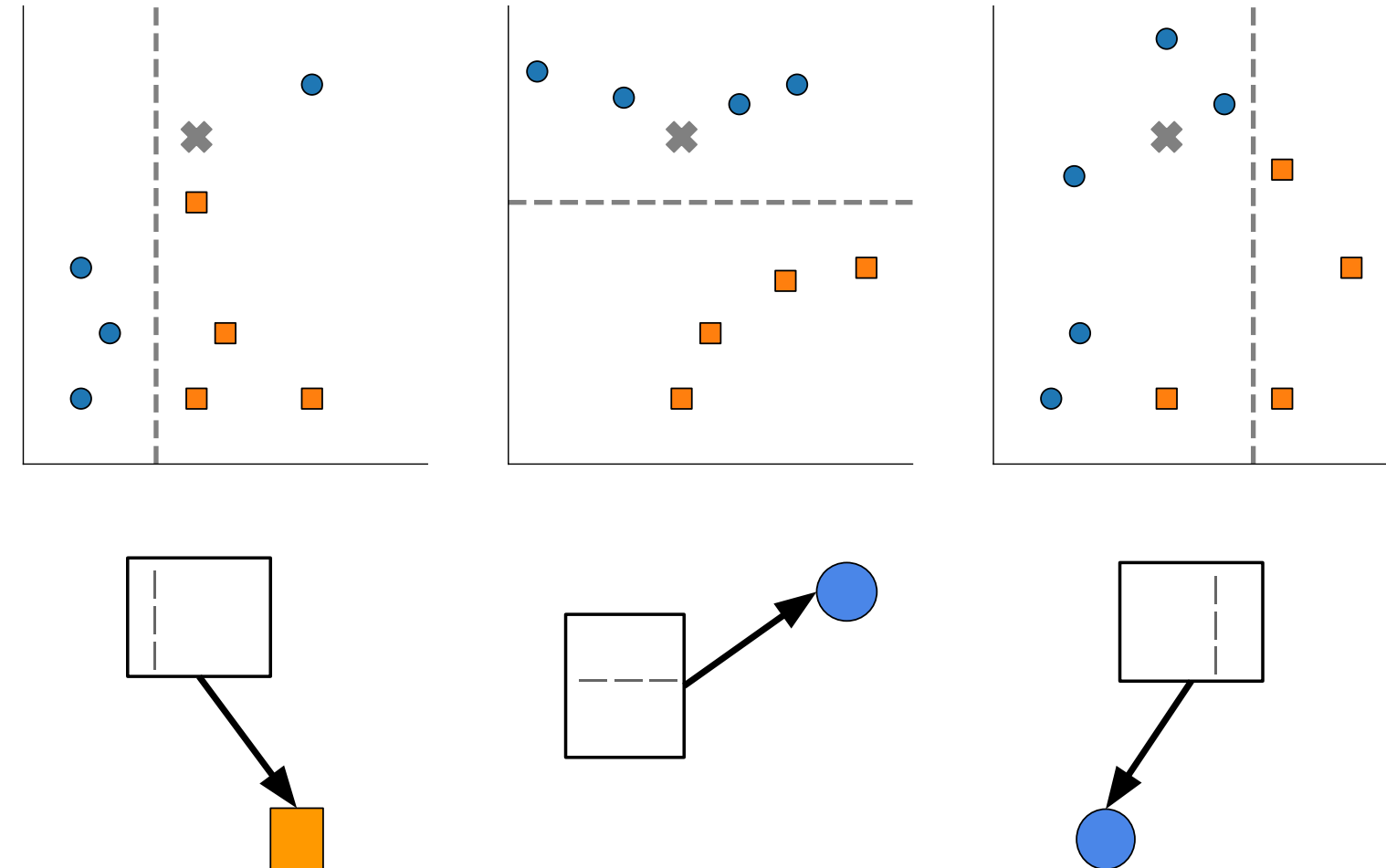
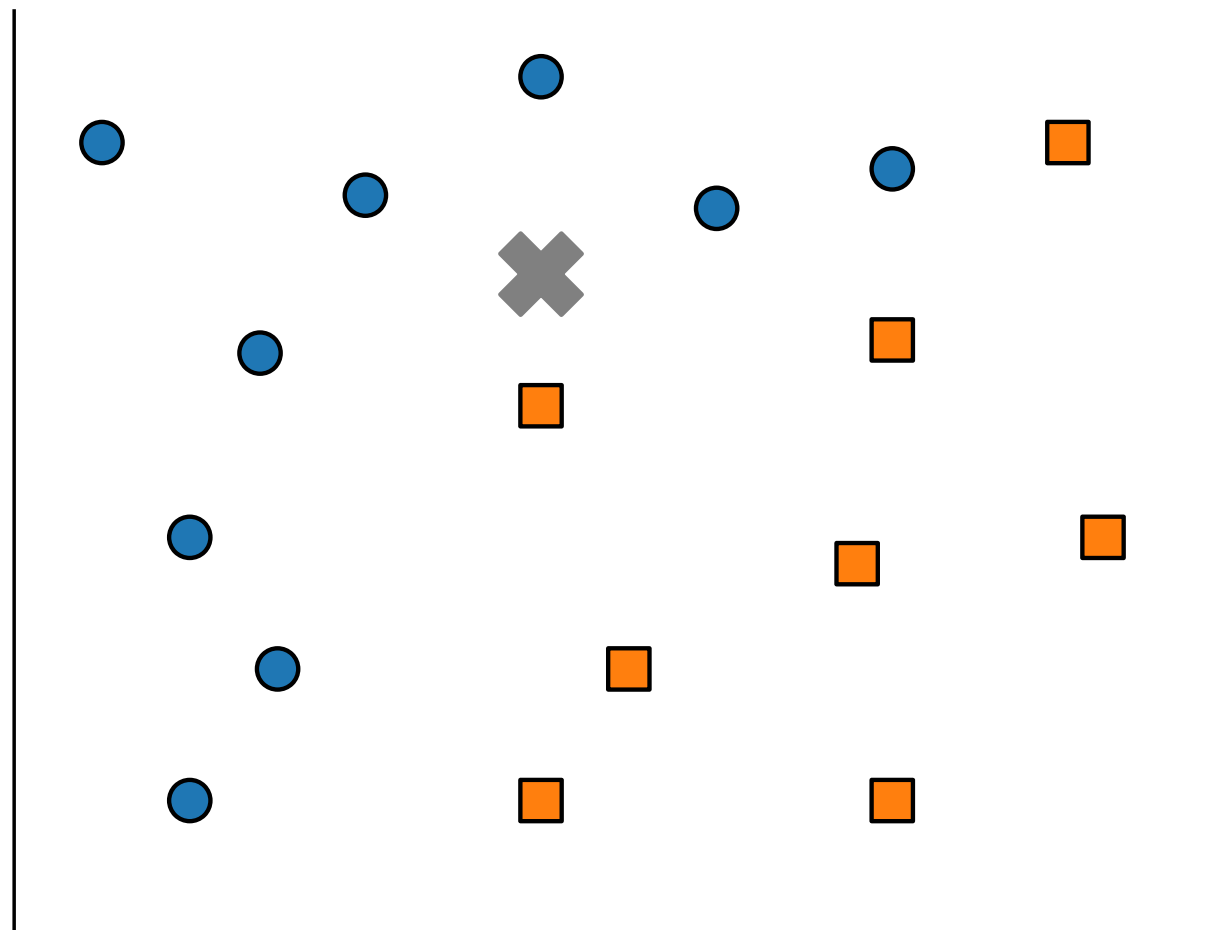


Bagging for classification

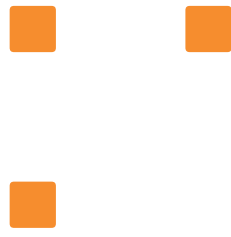




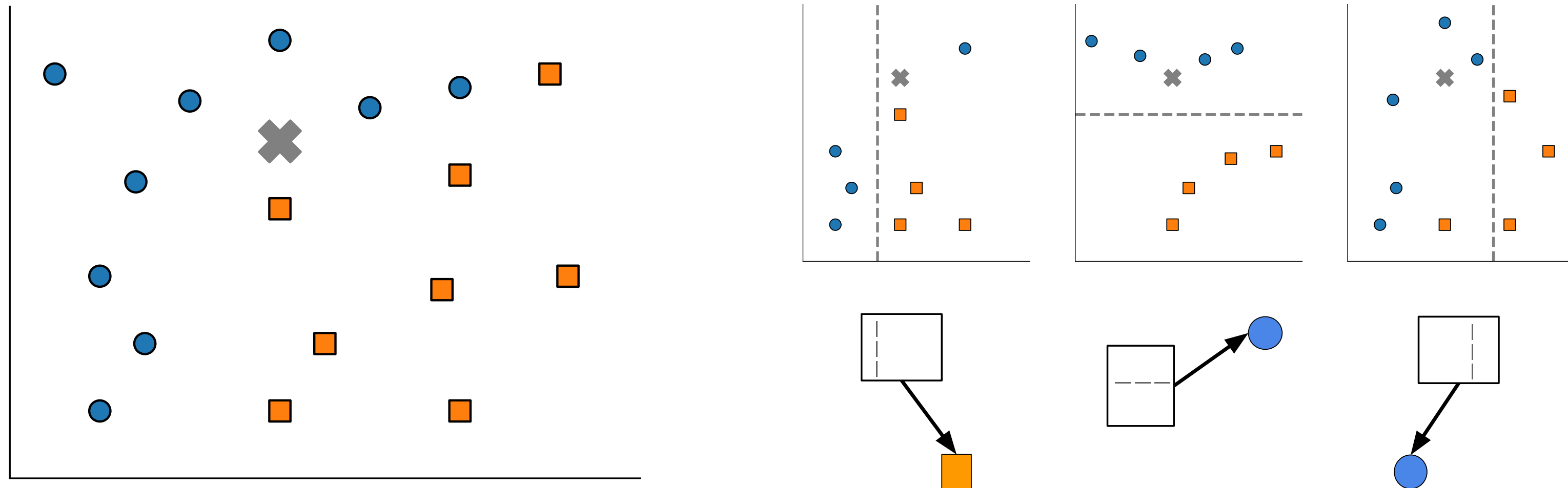
Bagging for classification



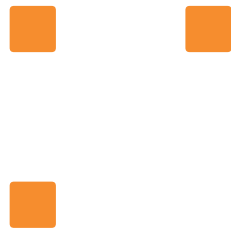
VOTE (, , ) = 



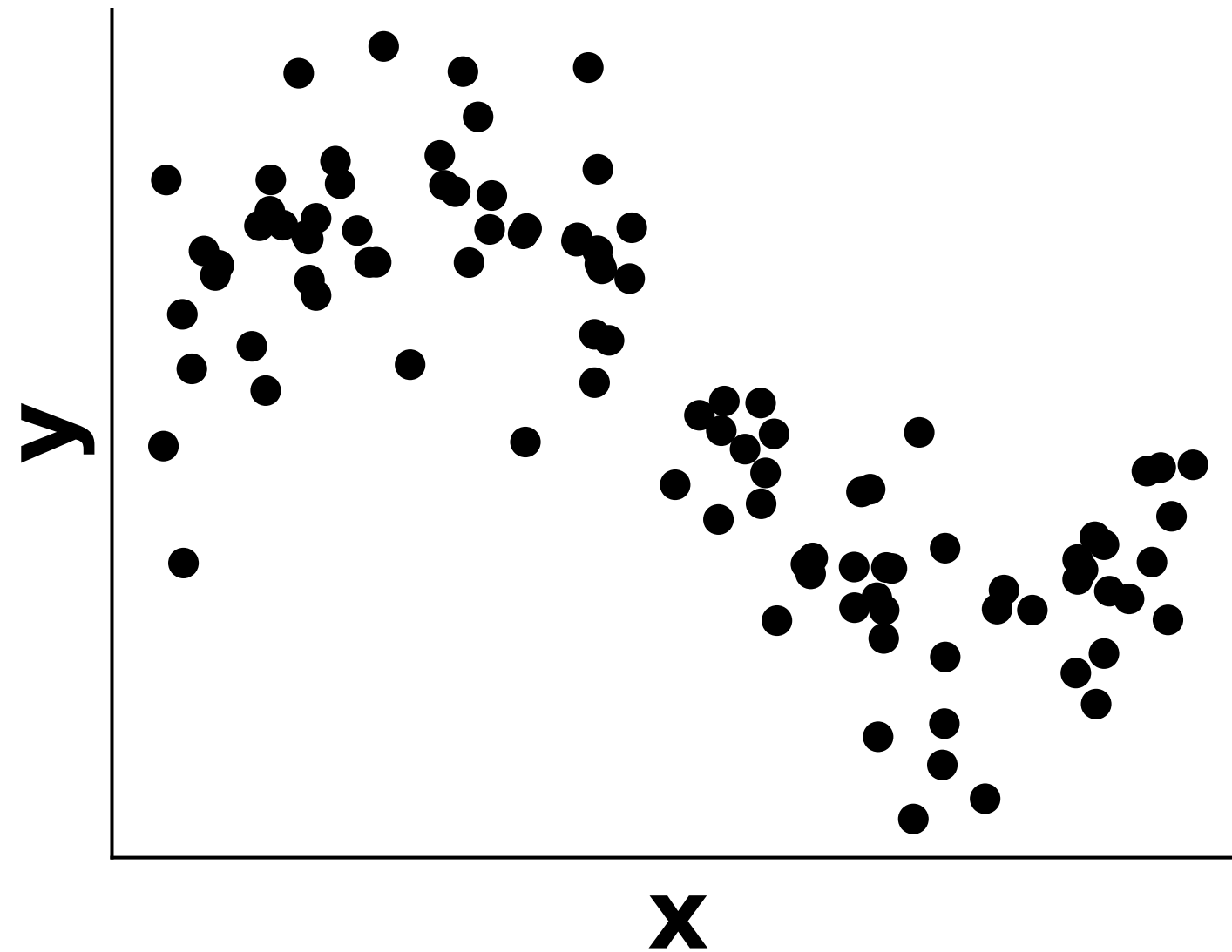
Bagging for classification



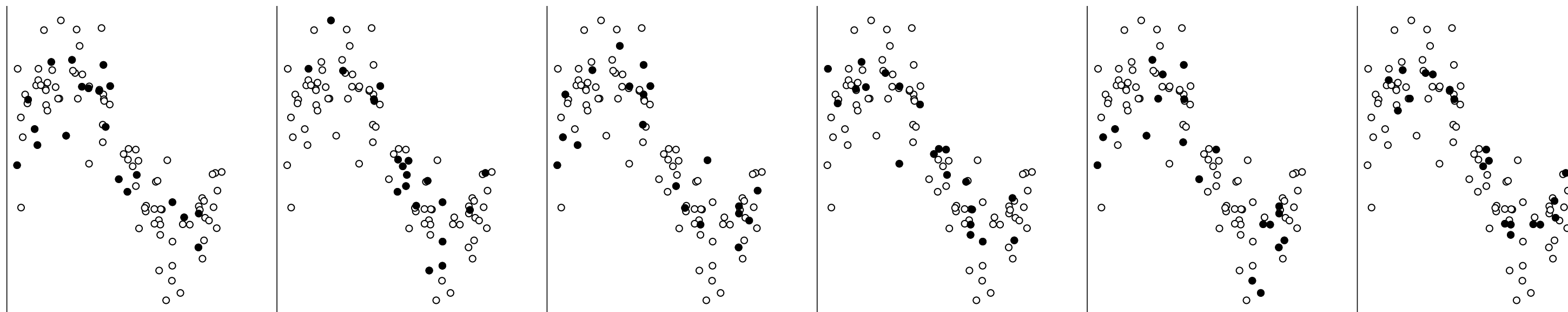
```
from sklearn.ensemble import BaggingClassifier  
from sklearn.ensemble import RandomForestClassifier
```



Bagging for regression

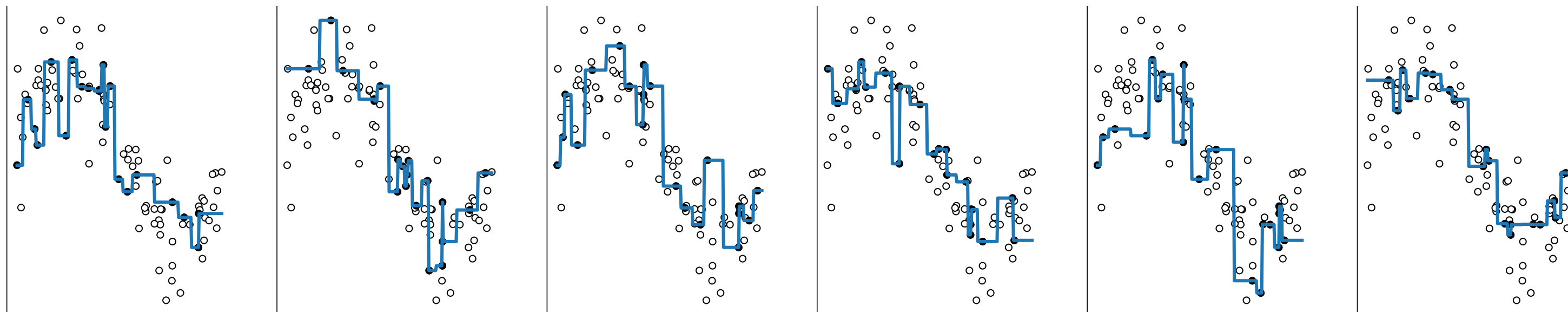


Bagging for regression



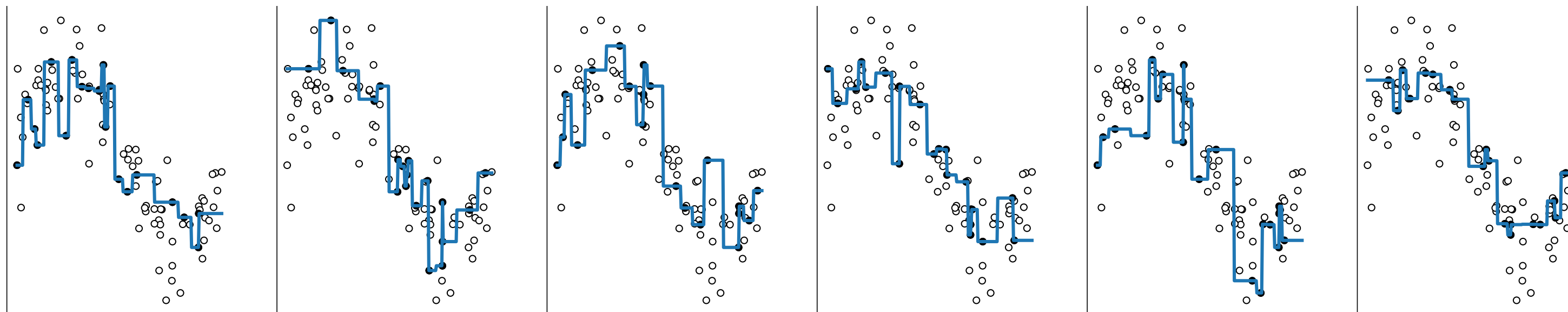
- Select multiple random subsets of the data

Bagging for regression

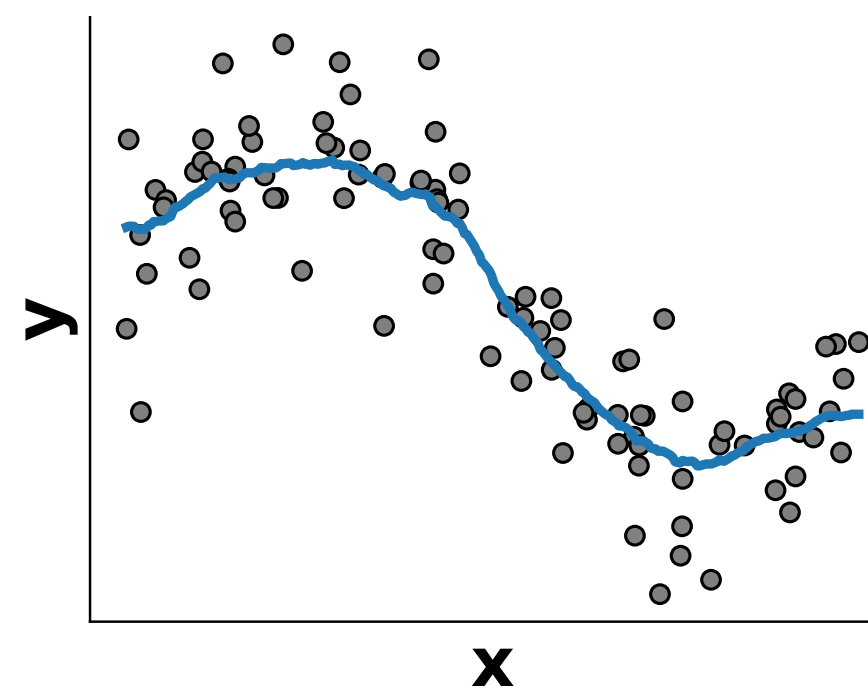


- Select multiple random subsets of the data
- Fit one model on each

Bagging for regression



- Select multiple random subsets of the data
- Fit one model on each
- Average predictions



Bagging versus Random Forests

Bagging is a general strategy

- Can work with any base model (linear, trees...)

Random Forests are bagged randomized decision trees

- At each split: a random subset of features are selected
- The best split is taken among the restricted subset
- Extra randomization decorrelates the prediction errors
- Uncorrelated errors make bagging work better

Take home messages on bagging

Bagging and random forests fit trees independently

- each deep tree overfits individually
- averaging the tree predictions reduces overfitting